EFREI Data Integration & Applications (ST2DLDI)
Data & Artificial Intelligence
June 26

TP : Open Data Profiling, Data Quality Analysis, Transformations, Modeling and Medallion Architecture

Dataset: French Road Safety Open Data (Accidents corporels : data.gouv.fr)

Jeu de données - Bases de données annuelles des accidents corporels de la circulation routière - Années de 2005 à 2024 | data.gouv.fr

Description of the dataset : https://www.data.gouv.fr/datasets/bases-de-donnees-annuelles-des-accidents-corporels-de-la-circulation-routiere-annees-de-2005-a-2024?resource_id=8ef4c2a3-91a0-4d98-ae3a-989bde87b62a

Part 1 : Data Profiling and Data Quality Analysis

Objective

Perform a complete profiling of the dataset and evaluate its data quality

Quantify missingness, detect anomalies, and propose remediation strategies

Instructions

You will work with the dataset provided (extracted from data.gouv.fr). Answer all guided questions clearly and justify your observations.

Guided Questions

A. Dataset Structure

- **Column inventory** :List all columns and their data types.
- **Semantic meaning** :Explain the meaning and purpose of each column based on the description provided.

B. Missing Values and Completeness

- **Missing percentages** :Compute the percentage of missing values for each column.
- **Critical missingness** :Identify which missing values are most problematic for analysis and why.
- **Remediation strategies** :Suggest appropriate actions.

C. Consistency and Validity Checks

- **Value ranges** :Identify out-of-range values or anomalies (Invalid coordinates, negative ages).
- **Categorical anomalies** :Detect unexpected or invalid categories.
- **Duplicates** :Identify and quantify duplicate records.

D. Data Quality Summary

- **Quality report** :Summarize the main issues discovered.
- **Impact analysis** :Explain how these issues affect downstream analytics.

Part 2 :Transformations, Modeling and Medallion Architecture

Objective

Design the transformation pipeline based on the profiling results, propose a simple analytical model, and produce a complete Medallion architecture diagram (Bronze → Silver → Gold).

A. Required Transformations (Silver Layer)

Based on the issues identified in Part 1, propose and justify the transformations needed :

- **Standardization** : Normalize formats (dates, coordinates, categories).
- **Cleaning** : Correct or remove invalid values.
- **Enrichment** : Add derived fields (e.g., accident severity index, time-of-day category).
- **Deduplication** : Remove duplicate records.
- **Documentation** : Document all transformation steps clearly.

B. Modeling (Gold Layer)

Design a simple analytical model to support road-safety analysis.

Suggested structure with fact table (accidents) + dimensions as location, vehicle etc.

C. Medallion Architecture Diagram

Produce a complete diagram showing:

- **Bronze:** raw open-data files
- **Silver:** cleaned, standardized, enriched tables
- **Gold:** analytical model (fact + dimensions)
- **BI / dashboards:** final consumption layer

Deliverables (End of Day)

1. Data Profiling Report
2. Transformation Plan (Silver layer)
3. Analytical Model (Gold layer)
4. Medallion Architecture Diagram
5. Short justification of design choices